Sebastian Tamayo

Senior Software Engineer

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Accomplished Senior Web Developer with over a decade of experience designing, developing, and optimizing scalable, high-performance web applications. Specializes in building intuitive, mobile-first interfaces with Vue.js and architecting secure and extensible backend systems using Python Django, Flask, FastAPI and RESTful APIs.

Skilled across the full software development lifecycle—from system architecture and relational/NoSQL data modeling to CI/CD automation, testing strategies, and cloud-native deployments on AWS and Kubernetes. Thrives in Agile, cross-functional teams, consistently delivering user-centered, business-aligned solutions through rapid iteration and stakeholder collaboration.

Renowned for writing clean, maintainable, and testable code while mentoring engineering teams and promoting best practices rooted in SOLID principles, clean architecture, and Test-Driven Development (TDD). Adept in frontend performance tuning, modular Vue component design, and building scalable backend services with Django and Celery.

Driven by a passion for innovation and continuous learning, with deep expertise in modern JavaScript frameworks, the Python ecosystem, and DevOps tooling. Committed to delivering impactful, real-world software solutions that enhance user experience and create measurable business value.

PROFESSIONAL EXPERIENCE

UST

Senior Software Engineer

June 2022 – June 2025 (3 years)

- Collaborated within a cross-functional Agile Scrum team, mentoring junior engineers and conducting rigorous code reviews focused on SOLID principles, design patterns, and clean architecture, ensuring long-term scalability and maintainability across frontend and backend codebases.
- Revamped a high-traffic, customer-facing application by developing a Vue.js frontend with responsive design, WCAG-compliant accessibility, and real-time user interactions. Integrated live auction streaming via AWS IVS and third-party systems like the Shopify Storefront API, increases in engagement and conversion rates.
- Engineered scalable backend services using Python Django and Django REST Framework (DRF), Flask, FastAPI implementing RESTful APIs and integrating with PostgreSQL for efficient ORM-based data access, transactional consistency, and normalized schema design.
- Architected a modular microservices-inspired backend deployed on Amazon EKS (Kubernetes). Leveraged Celery with Redis and Django Channels to support asynchronous processing and real-time communication between distributed Vue frontends and backend services, enhancing responsiveness and scalability.
- Led the development of an AI-powered chatbot using AWS Lex, AWS Lambda, and Amazon DynamoDB, enabling intelligent, context-aware support automation that scaled dynamically with user demand.
- Optimized frontend performance through Vue lazy loading, dynamic imports, and component-level memoization. On the backend, improved Django performance using async views, query optimizations, and Redis caching, leading to a 30% reduction in load time and enhanced user experience.
- Built and maintained robust CI/CD pipelines using GitHub Actions and Jenkins, integrating automated tests via Pytest, Vue Test Utils, and Cypress for E2E flows. This resulted in improved test coverage, faster feedback loops, and a 22% decrease in deployment cycle time.

Tech Stack: Vue.js, Vuex, TypeScript, HTML5, SCSS, Python, Django, Django REST Framework (DRF), Django Channels, Flask, FastAPI, PostgreSQL, Redis, DynamoDB, Celery, WebSockets, Shopify Storefront API, AWS (EKS, Lambda, IVS, Lex, S3, CloudWatch), GitHub Actions, Jenkins, Pytest, Vue Test Utils, Cypress, Webpack, Vite, WCAG, Figma, Postman, SEO, TDD

Gorilla Logic

Full Stack Engineer

February 2018 – May 2022 (4 years 4 months)

- Designed and implemented responsive, mobile-first interfaces using Vue.js, TypeScript, and SCSS, resulting in a 40% increase in mobile user retention and a 30% boost in overall platform engagement.
- Built scalable, reusable component libraries for enterprise dashboards, utilizing Pinia (or Vuex) for centralized state management, enabling consistency and efficiency across multiple frontend teams.
- Developed secure, high-performance backend services using Python Django and Django REST Framework (DRF), delivering clean, modular RESTful APIs while following domain-driven design and adhering to best practices in ORM modeling.
- Implemented real-time content delivery through Django Channels and Server-Sent Events (SSE), enabling dynamic UI updates and seamless interactivity for features such as live notifications and dashboards.
- Deployed containerized microservices using Docker on AWS ECS, and automated infrastructure provisioning with AWS CloudFormation, improving release velocity and ensuring environment consistency.
- Optimized backend performance through advanced PostgreSQL query tuning, schema normalization, and indexing strategies—reducing API response times by 15% under peak usage scenarios.
- Enabled asynchronous processing and decoupled communication patterns using Celery with Redis and optionally RabbitMQ, improving scalability, fault tolerance, and backend responsiveness.
- Practiced Test-Driven Development (TDD) using Pytest and Vue Test Utils, and established a robust observability stack with Prometheus, Grafana, Elastic APM, and AWS CloudWatch, facilitating proactive monitoring and rapid issue resolution.

Tech Stack: Vue.js, Pinia (Vuex), TypeScript, SCSS, Python, Django, Django REST Framework (DRF), Django Channels, Flask, FastAPI, Celery, PostgreSQL, Redis, RabbitMQ, WebSockets, REST API, Docker, AWS (ECS, Lambda, S3, CloudFormation, CloudWatch), Prometheus, Grafana, Elastic APM, Git, CI/CD, Pytest, Vue Test Utils, Postman, Figma, SEO, WCAG, TDD

Sophos Solutions

5 years 6 months

Senior Web Developer

September 2015 – February 2018 (2 years 6 months)

- Spearheaded the development of a high-performance Single Page Application (SPA) for a financial services platform using Vue.js, Composition API, and Pinia (or Vuex), leading to a 30% increase in user engagement and significantly faster client-side responsiveness.
- Modernized a legacy monolith by decoupling the frontend and integrating it with scalable RESTful APIs built using Python Django and Django REST Framework (DRF), reducing server response times by 40% and improving backend maintainability through modular service design.
- Boosted frontend performance through code splitting, lazy loading, tree shaking, and asset compression, cutting page load times by 35% and enhancing SEO rankings via structured metadata, Open Graph, and JSON-LD schema markup.
- Delivered fully responsive, mobile-first UIs aligned with WCAG 2.1 accessibility guidelines and optimized for Google AMP, resulting in a 25% growth in mobile traffic and better Lighthouse scores.
- Built a comprehensive testing strategy leveraging Vue Test Utils, Jest, Playwright, and Cypress, all integrated into GitHub Actions workflows, reducing post-deployment issues by 75% and increasing release confidence.
- Collaborated closely with UI/UX designers via Figma and Adobe XD to redesign and modularize complex enterprise dashboards, improving navigation performance by 50% and enhancing developer productivity across teams.

Tech Stack: Vue.js, Composition API, Pinia (Vuex), TypeScript, HTML5, SCSS, Python, Django, Django REST Framework (DRF), Flask, FastAPI, Celery, PostgreSQL, Redis, Vite, Webpack, GitHub Actions, Jenkins, Azure App Service, Playwright, Cypress, Jest, Vue Test Utils, Postman, WCAG, Google AMP, SEO, Open Graph, JSON-LD, Google Analytics, Adobe XD, CI/CD, TDD

Web Developer

September 2012 – February 2015 (2 years 6 months)

- Developed responsive, accessible, and standards-compliant web applications using Vue.js, TypeScript, HTML5, and SCSS, ensuring cross-browser compatibility, WCAG 2.1 compliance, and increased user engagement across diverse user groups.
- Built modular and reusable components leveraging Vue Composition API, Vue Router, and Pinia (or Vuex), enabling scalable and maintainable frontend architecture in line with evolving product requirements.
- Designed and integrated RESTful APIs using Python Django and Django REST Framework (DRF), delivering secure, extensible backend services and structured content models to support complex business workflows.
- Streamlined development workflows using Vite and Webpack on the frontend, and Django management commands and custom scripts on the backend, optimizing for build speed, asset handling, and cross-environment deployments.
- Collaborated with backend engineers to implement caching strategies using Redis, Memcached, and Django's low-level cache APIs, improving backend responsiveness and reducing server load for high-traffic endpoints.
- Tuned PostgreSQL performance by optimizing Django ORM queries, applying indexing strategies, and profiling bottlenecks using pg_stat_statements, which led to measurable improvements in backend throughput.
- Reduced frontend load times by 35% through lazy loading, code splitting, image optimization, and asset minification, improving SEO rankings and delivering smoother user experiences.

Tech Stack: Vue.js, TypeScript, HTML5, SCSS, Composition API, Vue Router, Pinia (Vuex), Python, Django, Django REST Framework (DRF), REST API, Flask, FastAPI, PostgreSQL, Redis, Memcached, Vite, Webpack, GitHub Actions, Jenkins, Azure App Service, Google Analytics, SEO, WCAG, CI/CD, Postman, Jest, Playwright, TDD, Figma

EDUCATION

Universidad Católica de Pereira 04.2005 – 09.2012

Bachelor of Computer Science

SKILLS

Languages:	C++, JS(ES6), TS, Python, C#, Rust, PHP, Java, HTML5
Frameworks and Libraries:	Vue.js, Vuex, Nuxt, React.js, Next.js, Gatesby, Angular, Node.js, Express, React Native, Redux, RxJS, REST API, Django, Flask, Fast API, JavaSpringBoot, Antd, GraphQL, Ably, Laravel, Tailwind CSS, RabbitMQ, Kafka, TypeORM, Shopify, WordPress
DBMS	MySQL, MS SQL, PostgreSQL, MongoDB, AWS DynamoDB, AWS RDS, Azure SQL Database
IDEs and Tools:	Visual Studio Code, Jenkins, Cypress, Jest, Jasmine, Terraform, Docker, Kubernetes, Redux DevTools, Babel, Webpack, Storybook, Azure DevOps, Git Bash, PowerShell
Cloud Services:	AWS Cloud Services, Microsoft Azure, GCP
Other skills:	Problem Solving, Full Stack Development, Critical Thinking, Leadership, Algorithm Design, Team-Oriented, SOLID, Agile (Scrum), Microservices, Test-Driven Development (TDD), Diagnosing, Business Analysts